

Network Options for Esports Event Venue

Introduction:

In an esports event venue, two primary networks are essential for ensuring a smooth experience: the **Esports Event LAN (wired)** and the **Pop-up Guest Wi-Fi (wireless)**.

The LAN network is designed for reliable, low-latency connections needed for competitive gaming, while the Wi-Fi network caters to casual visitors and participants who require internet access but don't necessarily need the high-performance bandwidth reserved for gamers.

Uses and Features:

1. Esports Event LAN (Wired):

Primary Use: Dedicated for competitive gameplay, ensuring the lowest possible latency and the most stable connection for esports players

Key Features:

Low Latency: Wired connections provide minimal delay, crucial for competitive gaming where every millisecond counts.

High Bandwidth: Supports large amounts of data transfer simultaneously without congestion, perfect for intensive multiplayer games.

Reliability: Physical connections are less susceptible to interference, ensuring consistent performance.

Security: Hardwired connections are more secure from external threats compared to wireless networks.

2. Pop-up Guest Wi-Fi (Wireless):

Primary Use: Designed for casual visitors, vendors, and spectators to access the internet while attending the event.

Key Features:

Convenience: Quick setup without extensive cabling.

Mobility: Guests can move around freely.

Moderate Bandwidth: Good for browsing, streaming, and social media.

Scalability: Supports many devices, though performance may decrease with crowd size.

Security: WPA2/WPA3 encryption, but more vulnerable than wired.

User Experience Review

- **Esports Event LAN (Wired):**

- Instant response times for players.

- No interruptions or lag spikes.

- Smooth streaming and in-game communications.

- **Pop-up Guest Wi-Fi (Wireless):**

- Visitors can browse, check scores, or post online.

- Connection quality depends on crowd density and distance.

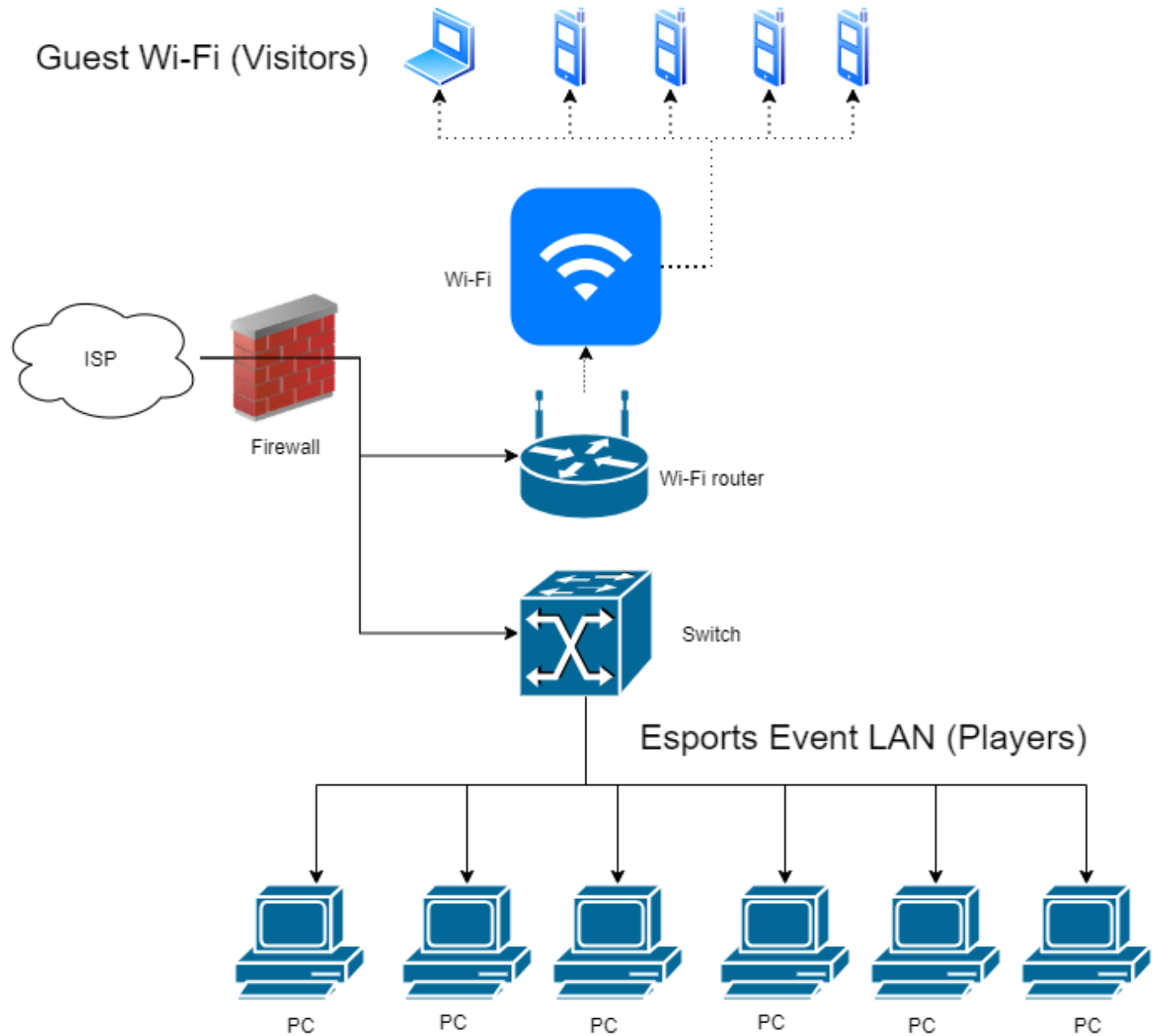
- Not suitable for competitive gaming.

Strengths and Weaknesses

- Esports Event LAN (Wired):
 - **Pros:** Low latency, reliable, secure, high bandwidth.
 - **Cons:** Expensive, less flexible, limited mobility.
- Pop-up Guest Wi-Fi (Wireless):
 - **Pros:** Easy setup, mobile, supports many devices, convenient for guests.
 - **Cons:** Higher latency, interference issues, performance drops with more users, less secure.

Diagram

This diagram shows the network setup: a wired LAN for players, providing stable low-latency connections, and a guest Wi-Fi for visitors, offering convenient internet access — both protected by a firewall and connected to the internet via the same ISP.



Verdict / Recommendation

The Esports Event LAN (Wired) is the best option for competitive match play because it provides low latency, high reliability, and secure performance.

The Pop-up Guest Wi-Fi (Wireless) should only be used as a **support network** for visitors, vendors, and spectators who need general internet access.